Google Colab Lab Assignment - Pretrained Model

Course Name: Deep Learning

Lab Title: Research Paper Implementation with Pre-trained Model (EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks)

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Reference Paper

Title: "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks"

Authors: Mingxing Tan and Quoc V. Le

Paper Overview

The research paper introduces EfficientNet, a convolutional neural network architecture designed to improve image classification performance while maintaining computational efficiency.

Traditional CNN models increase performance by scaling:

- **Depth** (more layers)
- **Width** (more channels)
- **Resolution** (larger images)

However, this paper proposes a **compound scaling method**, which balances all three factors simultaneously for better results.

EfficientNet models are pre-trained on large datasets like ImageNet and can be reused for different tasks using transfer learning.

What I Implemented

Applied the same transfer learning methodology using **MobileNetV2** (pre-trained on ImageNet) on the **Rice Leaf Disease Dataset** to classify 3 disease categories:

- Leaf Smut
- Bacterial Leaf Blight
- Brown Spot

Task 1: Research Paper Selection and Dataset Preparation

Instructions:

1. Select a research paper that applies a pre-trained model (e.g., VGG, ResNet, EfficientNet, etc.).
2. Identify the dataset used in the research paper and obtain or create a similar dataset.
   - **Dataset:** Rice Leaf Disease Dataset
   - **Dataset Link:** (https://www.kaggle.com/datasets/vbookshelf/rice-leaf-diseases)
   - **Description:** Contains images of rice leaves categorized into 3 disease classes: Leaf Smut, Bacterial Leaf Blight, and Brown Spot.
3. Perform necessary preprocessing steps:
   - Resize images to model input dimensions (224x224).
   - Apply MobileNetV2 preprocessing function.
4. Split the dataset into training (80%) and validation (20%) sets.

```
import zipfile

zip_path = "/content/RiceLeafDiseasesDataset.zip"

with zipfile.ZipFile(zip_path, 'r') as zip_ref:
    zip_ref.extractall("/content/dataset")

print("Dataset Extracted Successfully")
```

```
Dataset Extracted Successfully
```

```
import os

# Check top-level extracted folder
print("Top-level folders:", os.listdir("/content/dataset"))

# Check class folders
print("Classes:", os.listdir("/content/dataset/rice_leaf_diseases"))
```

```
Top-level folders: ['rice_leaf_diseases']
Classes: ['Bacterial leaf blight', 'Leaf smut', 'Brown spot']
```

```
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.applications.mobilenet_v2 import preprocess_input

DATASET_PATH = "/content/dataset/rice_leaf_diseases"
IMG_SIZE = 224
BATCH_SIZE = 32

# Apply MobileNetV2 preprocessing and 80/20 train-validation split
datagen = ImageDataGenerator(
    preprocessing_function=preprocess_input,
    validation_split=0.2
)

# Training set (80%)
train_data = datagen.flow_from_directory(
    DATASET_PATH,
    target_size=(IMG_SIZE, IMG_SIZE),
    batch_size=BATCH_SIZE,
    class_mode='categorical',
    subset='training'
)

# Validation set (20%)
val_data = datagen.flow_from_directory(
    DATASET_PATH,
    target_size=(IMG_SIZE, IMG_SIZE),
    batch_size=BATCH_SIZE,
    class_mode='categorical',
    subset='validation'
)

print(f"Training samples  : {train_data.samples}")
print(f"Validation samples: {val_data.samples}")
print(f"Classes           : {train_data.class_indices}")
```

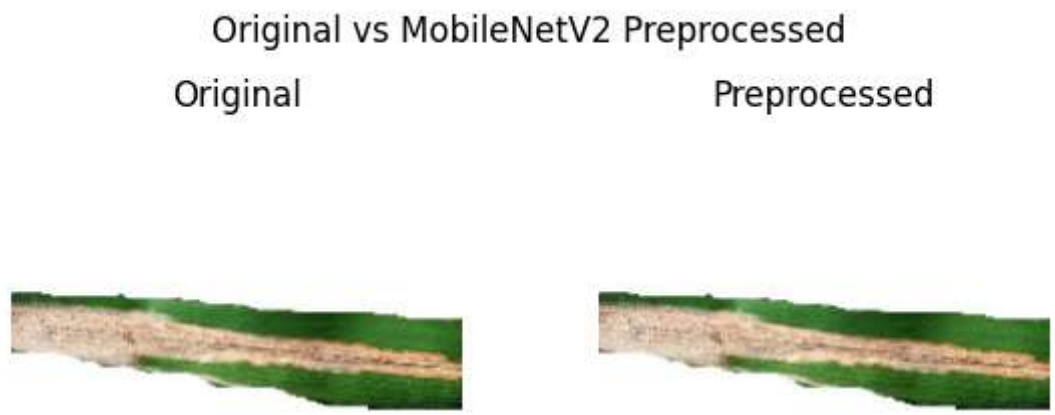
```
Found 96 images belonging to 3 classes.
Found 24 images belonging to 3 classes.
Training samples  : 96
Validation samples: 24
Classes           : {'Bacterial leaf blight': 0, 'Brown spot': 1, 'Leaf smut': 2
```

```
# ------------------------------------------------
# Visualization - Sample Images from Each Class
# ------------------------------------------------
import matplotlib.pyplot as plt
import numpy as np
```

```
# Get one batch of training images
sample_images, sample_labels = next(train_data)
class_names = list(train_data.class_indices.keys())

# --- Plot 1: Sample images from each class ---
plt.figure(figsize=(10, 4))
plt.suptitle("Sample Images from Each Class")

for i in range(min(5, len(sample_images))):
    plt.subplot(1, 5, i+1)
    # Denormalize for display (MobileNetV2 preprocessing scales to [-1, 1])
    img = (sample_images[i] + 1) / 2.0
    img = np.clip(img, 0, 1)
    plt.imshow(img)
    label_idx = np.argmax(sample_labels[i])
    plt.title(class_names[label_idx], fontsize=8)
    plt.axis('off')

plt.tight_layout()
plt.show()
```

```
# --- Plot 2: Original vs Preprocessed comparison ---
from tensorflow.keras.applications.mobilenet_v2 import preprocess_input
from tensorflow.keras.preprocessing.image import load_img, img_to_array
import os

# Load a raw image directly (before preprocessing)
class_folder = os.path.join(DATASET_PATH, class_names[0])
img_file = os.listdir(class_folder)[0]
raw_img = load_img(os.path.join(class_folder, img_file), target_size=(224, 224)
raw_array = img_to_array(raw_img) / 255.0  # just scaled to [0,1]

# Preprocessed version (MobileNetV2)
preprocessed = preprocess_input(img_to_array(raw_img))
preprocessed_display = (preprocessed + 1) / 2.0  # scale back for display

plt.figure(figsize=(6, 3))
```

```
plt.suptitle("Original vs MobileNetV2 Preprocessed")

plt.subplot(1, 2, 1)
plt.imshow(raw_array)
plt.title("Original")
plt.axis('off')

plt.subplot(1, 2, 2)
plt.imshow(np.clip(preprocessed_display, 0, 1))
plt.title("Preprocessed")
plt.axis('off')

plt.tight_layout()
plt.show()
```



Task 2: Model Implementation and Fine-tuning

Instructions:

1. Implement the pre-trained model as described in the research paper.
2. Visualize feature maps of a few layers.
3. Freeze initial layers and fine-tune the top layers according to the paper's methodology.
4. Optimize hyperparameters such as:
    - Learning rate: `0.00001`
    - Batch size: `32`
    - Number of epochs: `8`
    - Optimizer: `Adam`

5. Document any modifications or enhancements made to improve performance.

```
from tensorflow.keras.applications import MobileNetV2
from tensorflow.keras import layers, models

# Load MobileNetV2 pre-trained on ImageNet (without top classification layer)
base_model = MobileNetV2(
    weights='imagenet',
    include_top=False,
    input_shape=(224, 224, 3)
)

# Freeze all base model layers (feature extractor stays fixed)
for layer in base_model.layers:
    layer.trainable = False

print(f"Total base model layers (frozen): {len(base_model.layers)}")

# Add custom classification head on top
x = base_model.output
x = layers.GlobalAveragePooling2D()(x)
x = layers.Dense(128, activation='relu')(x)
output = layers.Dense(train_data.num_classes, activation='softmax')(x)

model = models.Model(inputs=base_model.input, outputs=output)
model.summary()
```

```
Downloading data from https://storage.googleapis.com/tensorflow/keras-applicat
9406464/9406464 ━━━━━━━━━━━━━━━━━━━━ 1s 0us/step
Total base model layers (frozen): 154
Model: "functional"
```

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 224, 224, 3)	0	-
Conv1 (Conv2D)	(None, 112, 112, 32)	864	input_layer[0][0]
bn_Conv1 (BatchNormalizatio…	(None, 112, 112, 32)	128	Conv1[0][0]
Conv1_relu (ReLU)	(None, 112, 112, 32)	0	bn_Conv1[0][0]
expanded_conv_dept… (DepthwiseConv2D)	(None, 112, 112, 32)	288	Conv1_relu[0][0]
expanded_conv_dept… (BatchNormalizatio…	(None, 112, 112, 32)	128	expanded_conv_de…
expanded_conv_dept… (ReLU)	(None, 112, 112, 32)	0	expanded_conv_de…
expanded_conv_proj… (Conv2D)	(None, 112, 112, 16)	512	expanded_conv_de…
expanded_conv_proj… (BatchNormalizatio…	(None, 112, 112, 16)	64	expanded_conv_pr…
block_1_expand (Conv2D)	(None, 112, 112, 96)	1,536	expanded_conv_pr…
block_1_expand_BN (BatchNormalizatio…	(None, 112, 112, 96)	384	block_1_expand[0…
block_1_expand_relu (ReLU)	(None, 112, 112, 96)	0	block_1_expand_B…
block_1_pad (ZeroPadding2D)	(None, 113, 113, 96)	0	block_1_expand_r…
block_1_depthwise (DepthwiseConv2D)	(None, 56, 56, 96)	864	block_1_pad[0][0]
block_1_depthwise_… (BatchNormalizatio…	(None, 56, 56, 96)	384	block_1_depthwis…
block_1_depthwise_… (ReLU)	(None, 56, 56, 96)	0	block_1_depthwis…
block_1_project (Conv2D)	(None, 56, 56, 24)	2,304	block_1_depthwis…

block_1_project_BN (BatchNormalizatio…	(None, 56, 56, 24)	96	block_1_project[…
block_2_expand (Conv2D)	(None, 56, 56, 144)	3,456	block_1_project_…
block_2_expand_BN (BatchNormalizatio…	(None, 56, 56, 144)	576	block_2_expand[0…
block_2_expand_relu (ReLU)	(None, 56, 56, 144)	0	block_2_expand_B…
block_2_depthwise (DepthwiseConv2D)	(None, 56, 56, 144)	1,296	block_2_expand_r…
block_2_depthwise_… (BatchNormalizatio…	(None, 56, 56, 144)	576	block_2_depthwis…
block_2_depthwise_… (ReLU)	(None, 56, 56, 144)	0	block_2_depthwis…
block_2_project (Conv2D)	(None, 56, 56, 24)	3,456	block_2_depthwis…
block_2_project_BN (BatchNormalizatio…	(None, 56, 56, 24)	96	block_2_project[…
block_2_add (Add)	(None, 56, 56, 24)	0	block_1_project_… block_2_project_…
block_3_expand (Conv2D)	(None, 56, 56, 144)	3,456	block_2_add[0][0]
block_3_expand_BN (BatchNormalizatio…	(None, 56, 56, 144)	576	block_3_expand[0…
block_3_expand_relu (ReLU)	(None, 56, 56, 144)	0	block_3_expand_B…
block_3_pad (ZeroPadding2D)	(None, 57, 57, 144)	0	block_3_expand_r…
block_3_depthwise (DepthwiseConv2D)	(None, 28, 28, 144)	1,296	block_3_pad[0][0]
block_3_depthwise_… (BatchNormalizatio…	(None, 28, 28, 144)	576	block_3_depthwis…
block_3_depthwise_… (ReLU)	(None, 28, 28, 144)	0	block_3_depthwis…
block_3_project (Conv2D)	(None, 28, 28, 32)	4,608	block_3_depthwis…
block_3_project_BN (BatchNormalizatio…	(None, 28, 28, 32)	128	block_3_project[…

block_4_expand (Conv2D)	(None, 28, 28, 192)	6,144	block_3_project_…
block_4_expand_BN (BatchNormalizatio…	(None, 28, 28, 192)	768	block_4_expand[0…
block_4_expand_relu (ReLU)	(None, 28, 28, 192)	0	block_4_expand_B…
block_4_depthwise (DepthwiseConv2D)	(None, 28, 28, 192)	1,728	block_4_expand_r…
block_4_depthwise_… (BatchNormalizatio…	(None, 28, 28, 192)	768	block_4_depthwis…
block_4_depthwise_… (ReLU)	(None, 28, 28, 192)	0	block_4_depthwis…
block_4_project (Conv2D)	(None, 28, 28, 32)	6,144	block_4_depthwis…
block_4_project_BN (BatchNormalizatio…	(None, 28, 28, 32)	128	block_4_project[…
block_4_add (Add)	(None, 28, 28, 32)	0	block_3_project_… block_4_project_…
block_5_expand (Conv2D)	(None, 28, 28, 192)	6,144	block_4_add[0][0]
block_5_expand_BN (BatchNormalizatio…	(None, 28, 28, 192)	768	block_5_expand[0…
block_5_expand_relu (ReLU)	(None, 28, 28, 192)	0	block_5_expand_B…
block_5_depthwise (DepthwiseConv2D)	(None, 28, 28, 192)	1,728	block_5_expand_r…
block_5_depthwise_… (BatchNormalizatio…	(None, 28, 28, 192)	768	block_5_depthwis…
block_5_depthwise_… (ReLU)	(None, 28, 28, 192)	0	block_5_depthwis…
block_5_project (Conv2D)	(None, 28, 28, 32)	6,144	block_5_depthwis…
block_5_project_BN (BatchNormalizatio…	(None, 28, 28, 32)	128	block_5_project[…
block_5_add (Add)	(None, 28, 28, 32)	0	block_4_add[0][0… block_5_project_…
block_6_expand (Conv2D)	(None, 28, 28, 192)	6,144	block_5_add[0][0]
block_6_expand_BN	(None, 28, 28	768	block_6_expand[0

(BatchNormalizatio…	192)		
block_6_expand_relu (ReLU)	(None, 28, 28, 192)	0	block_6_expand_B…
block_6_pad (ZeroPadding2D)	(None, 29, 29, 192)	0	block_6_expand_r…
block_6_depthwise (DepthwiseConv2D)	(None, 14, 14, 192)	1,728	block_6_pad[0][0]
block_6_depthwise_… (BatchNormalizatio…	(None, 14, 14, 192)	768	block_6_depthwis…
block_6_depthwise_… (ReLU)	(None, 14, 14, 192)	0	block_6_depthwis…
block_6_project (Conv2D)	(None, 14, 14, 64)	12,288	block_6_depthwis…
block_6_project_BN (BatchNormalizatio…	(None, 14, 14, 64)	256	block_6_project[…
block_7_expand (Conv2D)	(None, 14, 14, 384)	24,576	block_6_project_…
block_7_expand_BN (BatchNormalizatio…	(None, 14, 14, 384)	1,536	block_7_expand[0…
block_7_expand_relu (ReLU)	(None, 14, 14, 384)	0	block_7_expand_B…
block_7_depthwise (DepthwiseConv2D)	(None, 14, 14, 384)	3,456	block_7_expand_r…
block_7_depthwise_… (BatchNormalizatio…	(None, 14, 14, 384)	1,536	block_7_depthwis…
block_7_depthwise_… (ReLU)	(None, 14, 14, 384)	0	block_7_depthwis…
block_7_project (Conv2D)	(None, 14, 14, 64)	24,576	block_7_depthwis…
block_7_project_BN (BatchNormalizatio…	(None, 14, 14, 64)	256	block_7_project[…
block_7_add (Add)	(None, 14, 14, 64)	0	block_6_project_… block_7_project_…
block_8_expand (Conv2D)	(None, 14, 14, 384)	24,576	block_7_add[0][0]
block_8_expand_BN (BatchNormalizatio…	(None, 14, 14, 384)	1,536	block_8_expand[0…
block_8_expand_relu	(None, 14, 14,	0	block_8_expand_B…

(ReLU)	384)		
block_8_depthwise (DepthwiseConv2D)	(None, 14, 14, 384)	3,456	block_8_expand_r…
block_8_depthwise_… (BatchNormalizatio…	(None, 14, 14, 384)	1,536	block_8_depthwis…
block_8_depthwise_… (ReLU)	(None, 14, 14, 384)	0	block_8_depthwis…
block_8_project (Conv2D)	(None, 14, 14, 64)	24,576	block_8_depthwis…
block_8_project_BN (BatchNormalizatio…	(None, 14, 14, 64)	256	block_8_project[…
block_8_add (Add)	(None, 14, 14, 64)	0	block_7_add[0][0… block_8_project_…
block_9_expand (Conv2D)	(None, 14, 14, 384)	24,576	block_8_add[0][0]
block_9_expand_BN (BatchNormalizatio…	(None, 14, 14, 384)	1,536	block_9_expand[0…
block_9_expand_relu (ReLU)	(None, 14, 14, 384)	0	block_9_expand_B…
block_9_depthwise (DepthwiseConv2D)	(None, 14, 14, 384)	3,456	block_9_expand_r…
block_9_depthwise_… (BatchNormalizatio…	(None, 14, 14, 384)	1,536	block_9_depthwis…
block_9_depthwise_… (ReLU)	(None, 14, 14, 384)	0	block_9_depthwis…
block_9_project (Conv2D)	(None, 14, 14, 64)	24,576	block_9_depthwis…
block_9_project_BN (BatchNormalizatio…	(None, 14, 14, 64)	256	block_9_project[…
block_9_add (Add)	(None, 14, 14, 64)	0	block_8_add[0][0… block_9_project_…
block_10_expand (Conv2D)	(None, 14, 14, 384)	24,576	block_9_add[0][0]
block_10_expand_BN (BatchNormalizatio…	(None, 14, 14, 384)	1,536	block_10_expand[…
block_10_expand_re… (ReLU)	(None, 14, 14, 384)	0	block_10_expand_…
block_10_depthwise (DepthwiseConv2D)	(None, 14, 14, 384)	3,456	block_10_expand_…

block_10_depthwise… (BatchNormalizatio…	(None, 14, 14, 384)	1,536	block_10_depthwi…
block_10_depthwise… (ReLU)	(None, 14, 14, 384)	0	block_10_depthwi…
block_10_project (Conv2D)	(None, 14, 14, 96)	36,864	block_10_depthwi…
block_10_project_BN (BatchNormalizatio…	(None, 14, 14, 96)	384	block_10_project…
block_11_expand (Conv2D)	(None, 14, 14, 576)	55,296	block_10_project…
block_11_expand_BN (BatchNormalizatio…	(None, 14, 14, 576)	2,304	block_11_expand[…
block_11_expand_re… (ReLU)	(None, 14, 14, 576)	0	block_11_expand_…
block_11_depthwise (DepthwiseConv2D)	(None, 14, 14, 576)	5,184	block_11_expand_…
block_11_depthwise… (BatchNormalizatio…	(None, 14, 14, 576)	2,304	block_11_depthwi…
block_11_depthwise… (ReLU)	(None, 14, 14, 576)	0	block_11_depthwi…
block_11_project (Conv2D)	(None, 14, 14, 96)	55,296	block_11_depthwi…
block_11_project_BN (BatchNormalizatio…	(None, 14, 14, 96)	384	block_11_project…
block_11_add (Add)	(None, 14, 14, 96)	0	block_10_project… block_11_project…
block_12_expand (Conv2D)	(None, 14, 14, 576)	55,296	block_11_add[0][…
block_12_expand_BN (BatchNormalizatio…	(None, 14, 14, 576)	2,304	block_12_expand[…
block_12_expand_re… (ReLU)	(None, 14, 14, 576)	0	block_12_expand_…
block_12_depthwise (DepthwiseConv2D)	(None, 14, 14, 576)	5,184	block_12_expand_…
block_12_depthwise… (BatchNormalizatio…	(None, 14, 14, 576)	2,304	block_12_depthwi…
block_12_depthwise… (ReLU)	(None, 14, 14, 576)	0	block_12_depthwi…